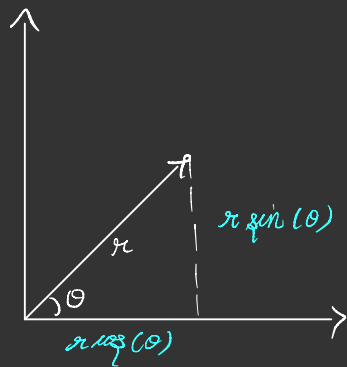


Polar coordinates: r, θ both vary w/ time



$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

$$\dot{x} = \dot{r} \cos(\theta) - r \sin(\theta) \dot{\theta}$$

$$\dot{y} = \dot{r} \sin(\theta) + r \cos(\theta) \dot{\theta}$$

$$\mathcal{L} = T = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$\mathcal{L} = \frac{1}{2} m \left[(\dot{r}^2 \cos^2(\theta) + r^2 \sin^2(\theta) \dot{\theta}^2 - \cancel{\dot{r} r \sin(2\theta) \dot{\theta}}) + (\dot{r}^2 \sin^2(\theta) + r^2 \cos^2(\theta) \dot{\theta}^2 + \cancel{\dot{r} r \sin(2\theta) \dot{\theta}}) \right]$$

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial \mathcal{L}}{\partial x_i} \Rightarrow \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{r}} = \frac{\partial \mathcal{L}}{\partial r} ; \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{\partial \mathcal{L}}{\partial \theta}$$

$$\Rightarrow \frac{d}{dt} m(\dot{r}) = \cancel{m(r \dot{\theta}^2)}$$

$$\ddot{r} = r \dot{\theta}^2$$

$$\Rightarrow \frac{d}{dt} m r^2 \dot{\theta} = 0$$

$$\Rightarrow m r^2 \dot{\theta} = \text{const.} = L (\text{ang. mom.})$$

↓

★ angular mom. (X)

angular momentum (✓)